

Notes: Antràs and Caballero (JPE, 2009)

Trade and Capital Flows: A Financial Frictions Perspective

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1 Big picture

- **Question.** Are trade integration and capital mobility substitutes or complements? The classical Heckscher–Ohlin–Mundell view says they are substitutes for capital-scarce economies: opening to trade reduces the incentive for capital to flow toward those economies. Antràs and Caballero show that this conclusion can reverse when countries differ in financial development.
- **Core contribution.** The paper builds a trade model with financial frictions. Countries differ in the quality of financial contracting, and sectors differ in how dependent they are on external finance. Once these two forms of heterogeneity are introduced, trade affects the return to capital through the sectoral allocation of production.
- **North versus South.** North is financially developed. South is financially underdeveloped. In South, the financially dependent sector cannot absorb enough capital because entrepreneurs face borrowing constraints. As a result, too much capital is pushed into the unconstrained sector.
- **Key mechanism.** Under autarky, financial underdevelopment depresses the relative price of the unconstrained good and depresses the rental rate of capital. Under free trade, South can specialize more in the unconstrained sector. This specialization raises the marginal product of capital in South.
- **Main reversal.** With high trade barriers, the rental rate of capital is lower in South than in North, so capital wants to leave South. With free trade, the rental rate of capital is higher in South than in North, so capital wants to flow into South.
- **Why this is not standard Heckscher–Ohlin.** In the standard model, trade integration moves goods prices in a way that reduces capital's return in a capital-scarce country. Here, trade integration relaxes a financial misallocation problem by moving labor and production toward the less financially constrained sector.
- **Important phrase.** Trade and capital flows are *complements* for financially underdeveloped countries: deeper trade integration raises the incentive for capital to flow into those countries.
- **Policy implication.** If global imbalances partly reflect financial underdevelopment, protectionism can backfire. Raising trade barriers can lower South's return to capital and increase capital outflows from South to North.
- **Sequencing implication.** The model suggests that developing economies may want to liberalize trade before fully liberalizing the capital account. Opening the capital account without trade liberalization may lead to capital flight.
- **Takeaway.** The interaction between trade and capital flows depends on why countries differ. If the central difference is capital scarcity, trade and capital mobility tend to substitute for each other. If the central difference is financial development, they can become complements.

2 Key objects and environment

2.1 Two countries and two sectors

The paper studies a world with two countries and two goods.

- **North** is financially developed.
- **South** is financially underdeveloped.
- **Sector 1** is financially dependent or financially constrained.
- **Sector 2** is financially unconstrained.

The relative price of good 2 in units of good 1 is denoted by

$$p \equiv \frac{P_2}{P_1}, \quad (1)$$

where good 1 is the numeraire. A low value of p means that the unconstrained good is cheap relative to the financially dependent good.

Economic interpretation. Financially weak economies overproduce the unconstrained good because capital cannot be efficiently deployed in the financially dependent sector. This depresses p under autarky.

2.2 Preferences

All agents have identical Cobb–Douglas preferences:

$$U = \left(\frac{C_1}{h}\right)^h \left(\frac{C_2}{1-h}\right)^{1-h}, \quad 0 < h < 1. \quad (2)$$

The parameter h is the expenditure share on good 1. The expenditure share on good 2 is $1 - h$.

Interpretation. Demand is standard and homothetic. The main force in the model is not unusual preferences; it is the interaction between sectoral financial constraints and production allocation.

2.3 Production

Both sectors use capital and labor. In the benchmark model, technologies are symmetric:

$$Y_i = ZK_i^\alpha L_i^{1-\alpha}, \quad i \in \{1, 2\}, \quad 0 < \alpha < 1. \quad (3)$$

Capital and labor are mobile across sectors within a country. Labor markets and goods markets are competitive. The rental rate of capital is denoted by δ , and the wage is denoted by w .

Interpretation. The benchmark model deliberately removes standard technological comparative advantage. If trade patterns arise, they arise from financial development, not from sectoral productivity or factor-intensity differences.

2.4 Entrepreneurs, rentiers, and the credit multiplier

There are two types of capitalists.

- **Entrepreneurs** know how to produce in the financially dependent sector 1.
- **Rentiers** own capital but cannot directly operate sector 1.

Let m be the share of capital held by entrepreneurs. Entrepreneurs can borrow, but only up to a multiple of their own capital. In this note, write the credit multiplier as $\theta > 1$:

$$I_i \leq \theta K_i. \quad (4)$$

The paper focuses on the case in which the borrowing constraint binds:

$$m\theta < h. \quad (5)$$

Interpretation. In the unconstrained first best, sector 1 would receive fraction h of capital. But if entrepreneurs own only fraction m of capital and can lever up only to θK_i , then sector 1 can receive at most $m\theta K$. The condition $m\theta < h$ means sector 1 receives too little capital.

2.5 Financial development and financial dependence

The model uses two different concepts.

- **Financial development** is country-level. A higher θ means better pledgeability, better contracting, or better ability to finance investment.
- **Financial dependence** is sector-level. Sector 1 requires external finance and entrepreneur-specific human capital. Sector 2 does not.

The model therefore combines a country difference with a sector difference. This is what creates a financially driven comparative advantage.

2.6 The main endogenous variables

The key prices are:

$$p = \text{relative price of good 2}, \quad w = \text{wage}, \quad \delta = \text{rental rate of capital}, \quad (6)$$

plus the shadow value of the borrowing constraint:

$$\lambda = \text{shadow value of relaxing the sector-1 borrowing constraint}. \quad (7)$$

The important rankings are:

$$p_{\text{aut}}^S < p_{\text{aut}}^N, \quad \delta_{\text{aut}}^S < \delta_{\text{aut}}^N, \quad \delta_{\text{open}}^S > \delta_{\text{open}}^N. \quad (8)$$

These inequalities summarize the paper's reversal: financial underdevelopment lowers the autarky

return to capital, but trade integration can raise the return enough to reverse the international ranking.

3 Models and theoretical specifications

3.1 The frictionless benchmark

If capital markets were perfect, the two sectors would be symmetric in the benchmark model. The relative price would be

$$p = 1, \quad (9)$$

with fraction h of capital and labor allocated to sector 1 and fraction $1 - h$ allocated to sector 2.

Interpretation. In the frictionless benchmark, there is no financial comparative advantage. Trade integration would not create the paper's key result because there is no financial misallocation to undo.

3.2 Closed economy with financial frictions

When the borrowing constraint binds, the amount of capital allocated to the financially dependent sector is

$$K_1 = m\theta K < hK, \quad (10)$$

and the unconstrained sector receives

$$K_2 = (1 - m\theta)K. \quad (11)$$

Labor remains allocated according to demand shares in the benchmark Cobb–Douglas case:

$$L_1 = hL, \quad L_2 = (1 - h)L. \quad (12)$$

Goods-market clearing and labor-market clearing imply the autarky relative price:

$$p_{\text{aut}} = \left[\frac{m\theta(1 - h)}{h(1 - m\theta)} \right]^\alpha < 1. \quad (13)$$

Interpretation. Because the constrained sector receives too little capital, the economy produces too much of the unconstrained good. The price of the unconstrained good is therefore depressed.

The rental rate in terms of good 2 equals the marginal product of capital in sector 2:

$$\frac{\delta}{p} = \alpha Z \left(\frac{(1 - m\theta)K}{(1 - h)L} \right)^{\alpha-1}. \quad (14)$$

Since p is also depressed when θ is low, the rental rate in terms of good 1 is depressed as well.

Economic message. Financial underdevelopment lowers the return to capital even though South is not less productive in the technological sense. The distortion is allocative: capital is stuck outside the financially dependent sector.

3.3 Entrepreneurial rents

Entrepreneurial capital earns more than rentier capital because entrepreneurs own the scarce input needed to operate sector 1. If λ is the multiplier on the financial constraint, the return to entrepreneurial capital is

$$R = \delta + \lambda\theta. \quad (15)$$

The shadow value λ is positive when the borrowing constraint binds.

Interpretation. Financial underdevelopment creates two different returns: a low market rental rate for ordinary capital and a high shadow value for entrepreneurial capital. The paper's capital-flow results are mainly about mobile rentier capital.

3.4 Closed economy comparative statics

An increase in θ means better financial contractibility. In the closed economy:

$$\frac{\partial p_{\text{aut}}}{\partial \theta} > 0, \quad \frac{\partial \delta_{\text{aut}}}{\partial \theta} > 0, \quad \frac{\partial \lambda_{\text{aut}}}{\partial \theta} < 0. \quad (16)$$

Interpretation. Better finance allows more capital into sector 1, reduces oversupply in sector 2, raises the price of the unconstrained good, raises the rental rate of capital, and reduces the scarcity rent of entrepreneurs.

3.5 Open economy: South as a small open economy

Now let South trade with a financially developed North. North is large, so South faces a world price approximately equal to North's autarky price:

$$p = p_{\text{aut}}^{\text{N}} > p_{\text{aut}}^{\text{S}}. \quad (17)$$

The constraint still pins down capital in sector 1:

$$K_1 = m\theta K. \quad (18)$$

But trade breaks the link between local demand and local production. Labor can now move according to world prices:

$$L_1(p) = \frac{m\theta L}{(1 - m\theta)p^{1/\alpha} + m\theta}. \quad (19)$$

Because $L_1(p)$ is decreasing in p , trade integration shifts labor away from the financially dependent sector and toward the unconstrained sector.

Interpretation. Trade lets South specialize in the sector in which its financial frictions are less damaging. This specialization is the central mechanism of the paper.

3.6 Open economy rental rate

With trade, the rental rate in terms of good 2 is

$$\frac{\delta(p)}{p} = \alpha Z \left(\left[(1 - m\theta) + m\theta p^{-1/\alpha} \right] \frac{K}{L} \right)^{\alpha-1}. \quad (20)$$

The rental rate in terms of good 1 is

$$\delta(p) = \alpha Z p \left(\left[(1 - m\theta) + m\theta p^{-1/\alpha} \right] \frac{K}{L} \right)^{\alpha-1}. \quad (21)$$

Since $\alpha - 1 < 0$, an increase in p reduces the expression inside the parentheses and raises $\delta(p)/p$. It also raises $\delta(p)$.

Economic message. Trade integration raises the Southern return to capital. It does so because the economy reallocates labor toward the unconstrained sector, lowering the capital-labor ratio there and raising capital's marginal product.

3.7 Free trade factor-price ranking

Under free trade, South specializes relatively more in the unconstrained sector. In the benchmark case, free trade implies

$$w^S < w^N, \quad \delta^S > \delta^N. \quad (22)$$

Interpretation. South has lower wages because financial frictions keep productivity and the effective allocation of capital depressed. But the same low wage makes capital valuable when goods prices are equalized through trade.

4 Capital mobility, trade costs, and robustness

4.1 Capital mobility with large trade frictions

If trade frictions are large, South behaves like the autarkic financial-friction economy. Since

$$\delta_{\text{aut}}^N > \delta_{\text{aut}}^S, \quad (23)$$

ordinary rentier capital has an incentive to flow from South to North.

Economic message. When South opens the capital account but not goods trade, capital flight can occur. The problem is not that South lacks savings; it is that its financial system misallocates capital and lowers the market return to capital.

4.2 Capital mobility with free trade

With free trade, the ranking reverses:

$$\delta_{\text{open}}^S > \delta_{\text{open}}^N. \quad (24)$$

The capital flow needed to equalize rental rates runs from North to South. In the benchmark model, the size of the capital flow from North to South is

$$\frac{F^{N \rightarrow S}}{K} = \frac{(h - m\theta^N)(\theta^N - \theta^S)}{\theta^N(1 - h)}. \quad (25)$$

Interpretation. The larger the financial-development gap, the larger the incentive for capital to move, but the direction depends on trade openness. Under free trade, capital moves into South because South can specialize in its financially less constrained sector.

4.3 The complementarity result

The paper's main complementarity can be written as

$$\frac{\partial \delta^S}{\partial \text{trade openness}} > 0. \quad (26)$$

Since capital moves toward higher returns, this implies

$$\text{more trade openness in South} \implies \text{stronger incentive for capital inflows into South.} \quad (27)$$

Contrast with Heckscher–Ohlin–Mundell. In the classical model, trade integration tends to reduce the incentive for capital to move to capital-scarce countries. Here, trade integration increases the incentive for capital to move to financially underdeveloped countries.

4.4 Intermediate trade costs

Let $\tau \in (0, 1)$ denote an iceberg trade cost on good 2. South then faces a lower effective relative price:

$$p^S = p^N(1 - \tau). \quad (28)$$

There is a threshold $\bar{\tau}$ such that

$$\tau < \bar{\tau} \implies \delta^S > \delta^N, \quad (29)$$

while

$$\tau > \bar{\tau} \implies \delta^S < \delta^N. \quad (30)$$

Interpretation. Trade costs continuously govern the direction of capital flows. Low trade costs make South attractive to capital. High trade costs make South unattractive to capital.

4.5 General model: anti-Stolper–Samuelson logic

The authors show that the mechanism is not an artifact of Cobb–Douglas technologies or equal factor intensities. In a general model with homothetic preferences and constant-returns technologies, if South's autarky relative price is lower than North's,

$$p_{\text{aut}}^S < p_{\text{aut}}^N, \quad (31)$$

then trade integration lowers the wage-rental ratio and raises the real rental rate of capital in South.

Why this is surprising. If South is capital scarce, the standard Stolper–Samuelson intuition might suggest that trade integration should lower the return to capital. The paper shows that financial frictions can dominate this force.

Mechanism. As p rises, labor moves toward the unconstrained sector. Capital in that sector becomes scarcer relative to labor, so its marginal product rises. This holds even when sectoral factor intensities are more general.

4.6 Comparative advantage in the general model

The general model delivers two important results.

First, holding factor abundance fixed, better financial contractibility raises the autarky relative price of the unconstrained good:

$$\frac{\partial p_{\text{aut}}}{\partial \theta} > 0. \quad (32)$$

Second, the effect of aggregate capital abundance depends on factor intensities and elasticities of substitution. The relevant condition is

$$\frac{\alpha_1}{(1 - \alpha_1)\sigma_1} - \frac{\alpha_2}{(1 - \alpha_2)\sigma_2} > 0, \quad (33)$$

where α_i is one minus the labor share in sector i and σ_i is the elasticity of substitution between capital and labor in sector i .

Empirical discipline. The authors point to evidence that financially developed countries export more in financially dependent industries. They also report that correlations between financial dependence and labor shares, elasticities of substitution, and the theory-relevant composite term are small in U.S. industry data.

4.7 Dynamic extension

The dynamic model introduces saving and capital accumulation. Agents are born and die at rate ϕ , save during life, and consume at death. Capital is the store of value.

The return to holding capital is

$$r_t^j = \frac{\delta_t^j}{(p_t^j)^{1-h}} + (1 - h) \frac{\dot{p}_t^j}{p_t^j}, \quad j \in \{N, S\}. \quad (34)$$

In steady state, prices are constant, so

$$r^j = \frac{\delta^j}{(p^j)^{1-h}}. \quad (35)$$

Interpretation. The dynamic model translates the static rental-rate result into an interest-rate and net-capital-flow result.

4.8 Steady-state interest rates

Under autarky, the steady-state interest rate is increasing in financial development. Thus,

$$r_{\text{aut}}^N > r_{\text{aut}}^S. \quad (36)$$

Under free trade, the ranking reverses:

$$r_{\text{open}}^S > r_{\text{open}}^N. \quad (37)$$

Economic message. The dynamic model preserves the core complementarity. Trade integration raises South's interest rate and can turn capital outflows into capital inflows.

4.9 Protectionism and global imbalances

With intermediate trade frictions, the Southern interest rate depends on the effective trade price $p^N(1 - \tau)$. Higher trade frictions lower South's interest rate and increase the interest-rate gap:

$$\frac{\partial(r^N - r^S)}{\partial\tau} > 0. \quad (38)$$

Therefore, if South is initially sending capital to North, protectionism can increase the capital outflow from South.

Policy message. Trade restrictions aimed at reducing North's trade deficits can backfire when the imbalances are rooted in financial frictions. Less trade openness makes South less attractive to capital and can increase South's net saving outflow.

5 Propositions, figures, and how to read them

5.1 Proposition 1: closed economy comparative statics

Read:

- Higher financial contractibility θ raises the relative price p of the unconstrained good.
- Higher θ raises the real rental rate of capital.
- Higher θ lowers the shadow value of entrepreneurial capital.

Economic message. Better financial markets reduce the sectoral misallocation of capital. This raises the market return to ordinary capital and reduces the scarcity rent earned by entrepreneurs.

5.2 Proposition 2: trade raises South's rental rate

Read:

- Trade integration exposes South to a higher relative price p for the unconstrained good.

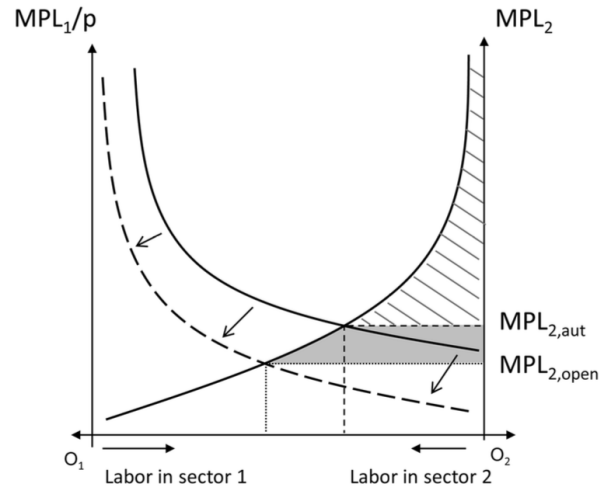


FIG. 1.—Trade integration and the rental rate of capital

- A higher p shifts labor toward the unconstrained sector.
- The marginal product of capital in that sector rises.

Economic message. Trade is valuable because it lets South reorganize production around the sector less damaged by financial frictions.

5.3 Figure 1: trade integration and the rental rate of capital

Read:

- The figure plots the value marginal product of labor in the two sectors.
- Trade integration raises p , shifting the relevant schedule and moving labor toward sector 2.
- The shaded area represents the increase in payments to capital in the unconstrained sector.

Economic message. The visual logic is simple: labor moves toward the unconstrained sector, capital there becomes relatively scarce, and the rental rate rises.

5.4 Proposition 3: free trade does not equalize factor prices

Read:

- South produces both goods and imports the financially dependent good 1.
- South exports the less financially dependent good 2.
- Wages remain lower in South, but the rental rate of capital is higher in South.

Economic message. Goods price equalization is not enough for factor price equalization when financial frictions constrain one sector.

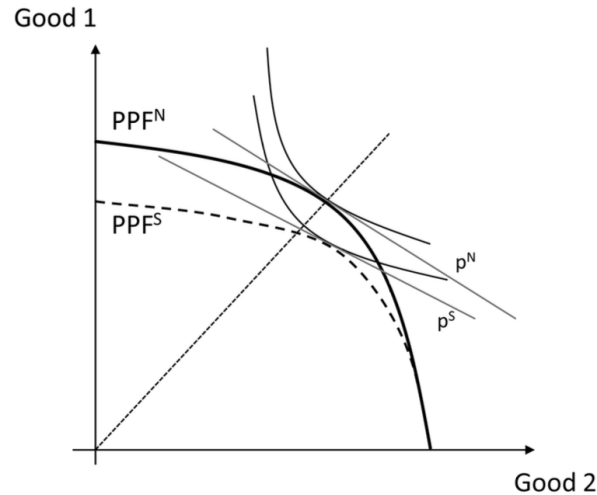


FIG. 2.—Effect of θ on autarky relative price p

5.5 Proposition 4: trade costs determine the direction of capital flows

Read:

- There is a threshold level of trade costs $\bar{\tau}$.
- Below the threshold, capital wants to flow from North to South.
- Above the threshold, capital wants to flow from South to North.

Economic message. Capital mobility and trade openness cannot be analyzed separately. The same financially underdeveloped country can export or import capital depending on how open goods markets are.

5.6 Proposition 5: general complementarity

Read:

- The complementarity result survives general homothetic preferences and general neoclassical technologies.
- The key condition is $p_{\text{aut}}^S < p_{\text{aut}}^N$.
- Under that condition, trade integration raises South's real rental rate of capital.

Economic message. The result does not rely on a special Cobb–Douglas trick. It relies on South having comparative advantage in the less financially constrained sector.

5.7 Figure 2: financial development and comparative advantage

Read:

- The figure compares North's and South's production possibility frontiers.
- Financial constraints bow in South's frontier in the region where sector 1 production is high.
- The slope of the frontier implies a lower autarky relative price p in South.

Economic message. Financial underdevelopment itself creates comparative advantage in the unconstrained sector.

5.8 Propositions 6 and 7: robustness to Heckscher–Ohlin forces

Read:

- Proposition 6 gives conditions under which financial development and factor abundance determine the autarky relative price.
- Proposition 7 states that when sectors differ only in financial dependence and North is at least as capital abundant as South, free trade leaves the rental rate higher in South than in North.

Economic message. Standard factor-abundance forces matter, but they do not automatically overturn the financial-frictions mechanism.

5.9 Propositions 8 and 9: dynamic capital flows and protectionism

Read:

- Proposition 8 says trade integration raises South's interest rate and reverses the autarky interest-rate ranking.
- Proposition 9 says that with trade frictions, there is a threshold determining whether savings flow to South or North.
- Higher protection raises the North–South interest-rate gap when South is already exporting capital.

Economic message. Protectionism can increase, not decrease, global imbalances if those imbalances are driven by financial frictions.

6 Short summary

- The paper studies how goods-market integration and capital mobility interact when countries differ in financial development.
- Sector 1 is financially dependent; sector 2 is financially unconstrained.
- In financially underdeveloped South, sector 1 cannot absorb enough capital, so sector 2 is oversupplied and its relative price is depressed.
- In autarky, South has a low rental rate of capital, so opening the capital account alone can induce capital outflows.

- Trade integration raises the relative price faced by South and shifts labor toward the unconstrained sector.
- This raises the marginal product of capital and the rental rate in South.
- With free trade, the rental rate and interest rate can be higher in South than in North, so capital flows into South.
- The result is the opposite of the classical Heckscher–Ohlin–Mundell substitutability result.
- The mechanism survives more general preferences, technologies, and dynamic capital accumulation.
- Protectionism can backfire by lowering South’s return to capital and increasing South-to-North capital flows.

7 Conclusion

7.1 Core conclusion

Antràs and Caballero show that trade and capital flows can be complements for financially underdeveloped economies. Trade integration raises the return to capital in South because it allows South to specialize in sectors that are less constrained by weak finance. This can turn capital outflows into capital inflows.

7.2 Economic intuition

Financial frictions misallocate capital. In autarky, local demand forces South to produce too much in the financially dependent sector despite weak finance, and the resulting allocation depresses the market return to capital. Trade breaks this link. South can import the financially dependent good and export the unconstrained good. Labor moves toward the unconstrained sector, capital becomes relatively scarce there, and the rental rate rises.

7.3 Why the paper matters

The paper provides a unified way to think about trade patterns, capital flows, and global imbalances. It shows that the standard trade-versus-capital-mobility logic depends on the source of international heterogeneity. If countries differ mainly in capital abundance, trade and capital mobility may be substitutes. If countries differ importantly in financial development, trade and capital mobility may be complements.

7.4 Takeaway

The key lesson is that financial frictions turn trade policy into capital-flow policy. For developing economies, deeper trade integration can make domestic capital more productive and attract foreign capital. For the world economy, protectionist responses to external imbalances can have the opposite of the intended effect.